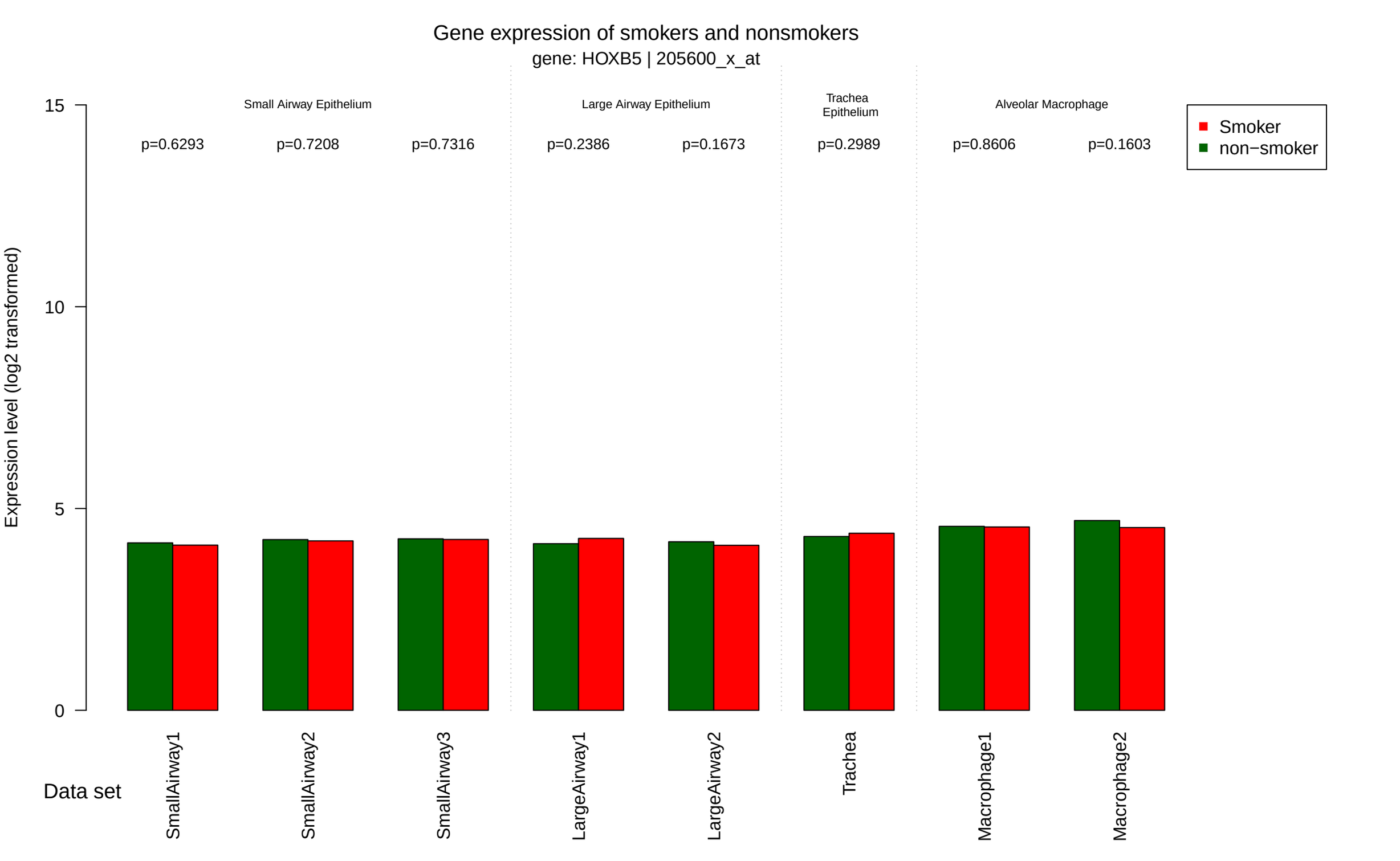
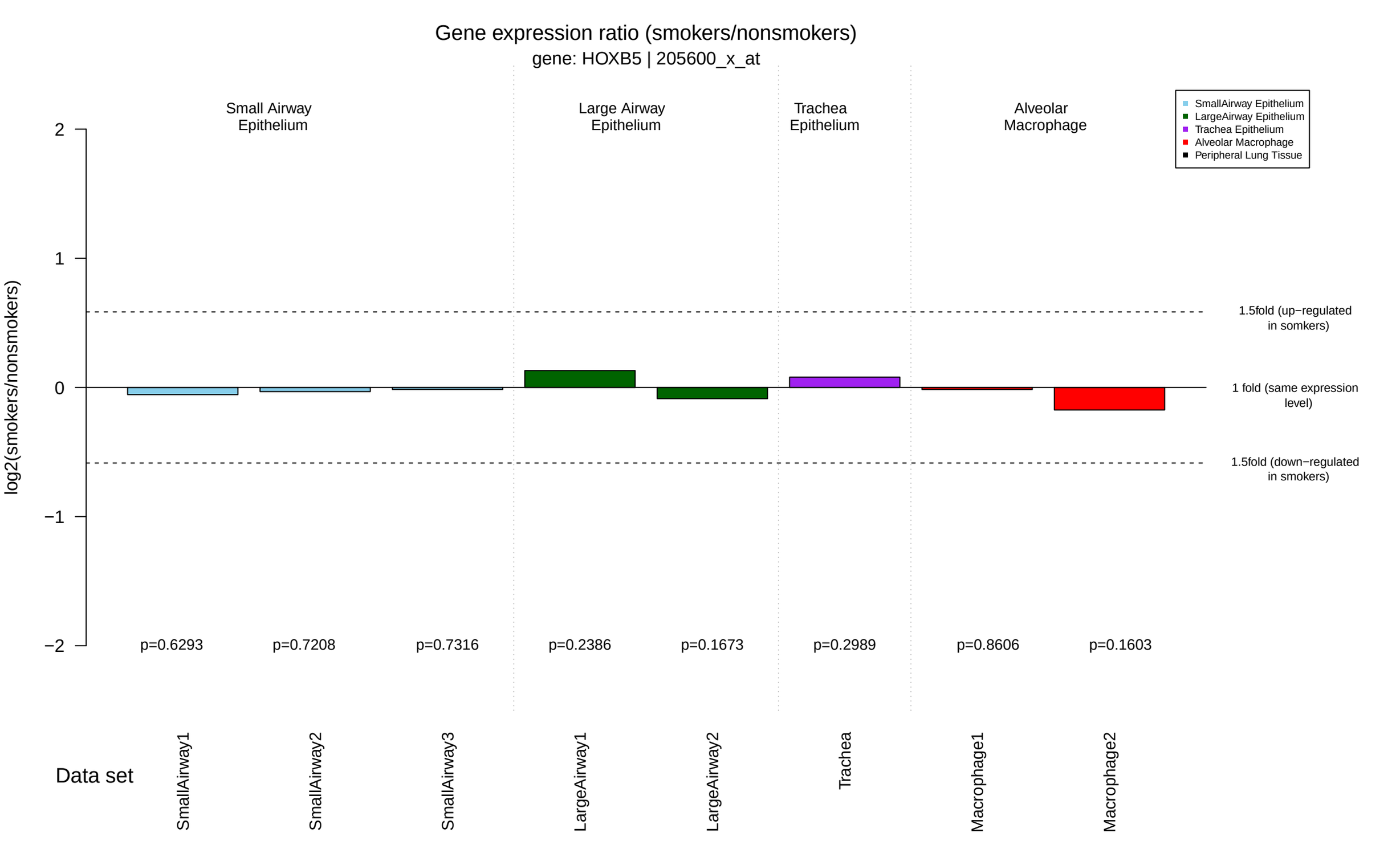
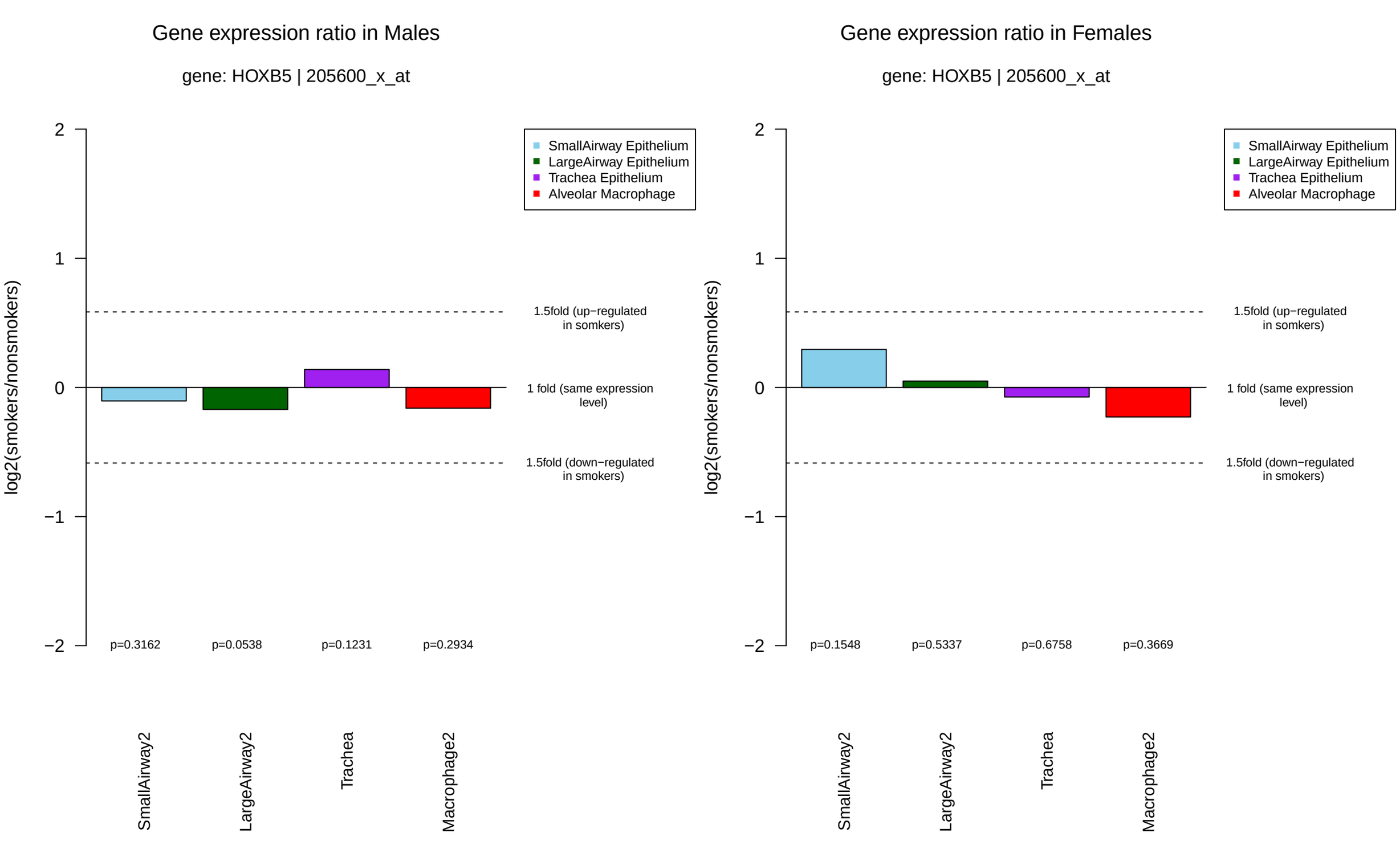
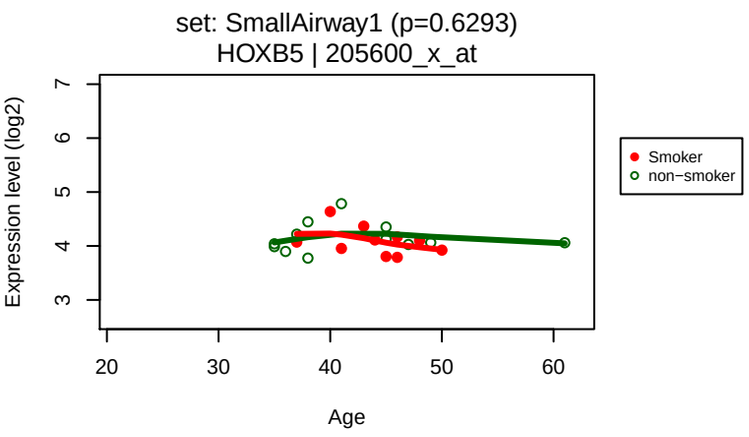
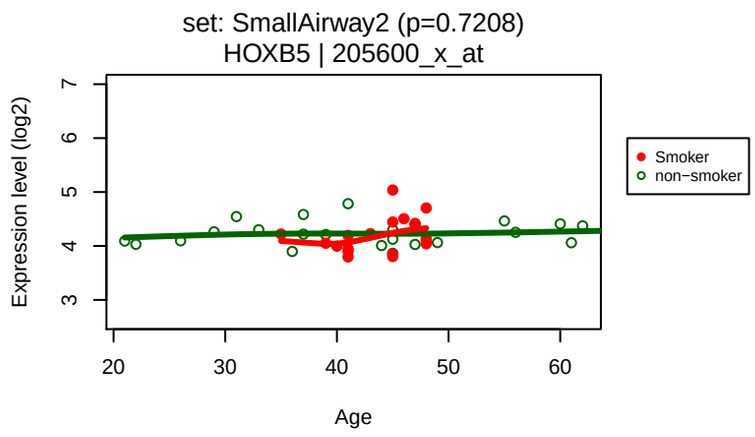
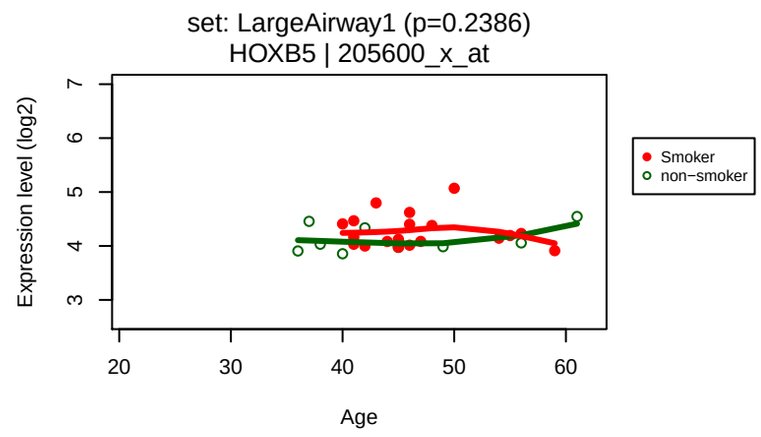

Gene expression of smokers and nonsmokers
gene: HOXB5 | 205600_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6293
p=0.7208
p=0.7316
p=0.2386
p=0.1673
p=0.2989
p=0.8606
p=0.1603
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

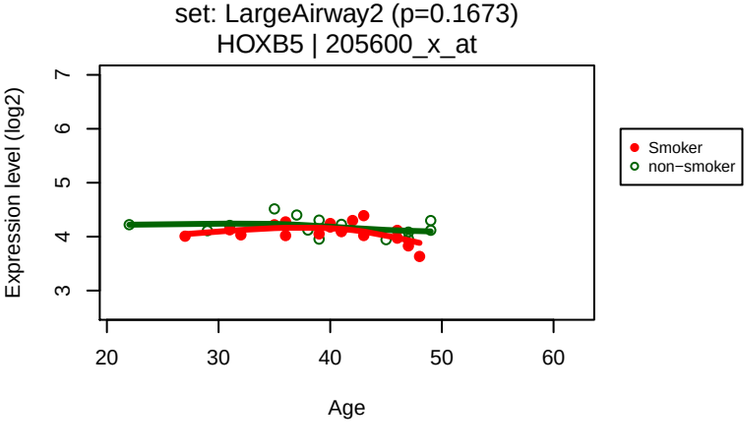

Gene expression ratio (smokers/nonsmokers)
gene: HOXB5 | 205600_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6293
p=0.7208
p=0.7316
p=0.2386
p=0.1673
p=0.2989
p=0.8606
p=0.1603
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

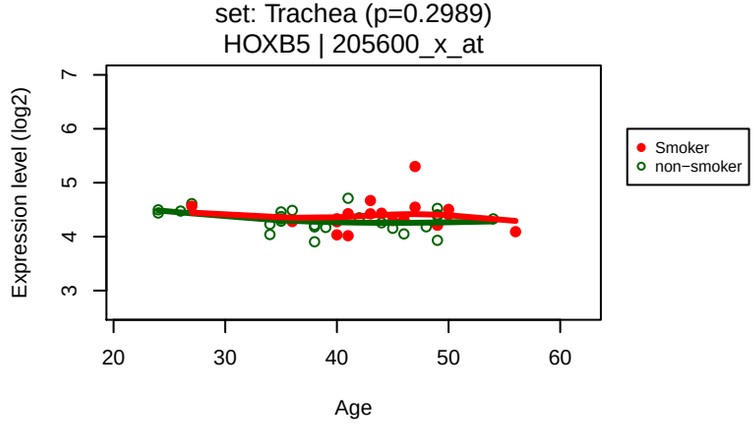

Gene expression ratio in Males
gene: HOXB5 | 205600_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3162
p=0.0538
p=0.1231
p=0.2934
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB5 | 205600_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1548
p=0.5337
p=0.6758
p=0.3669
SmallAirway2
LargeAirway2
Trachea
Macrophage2

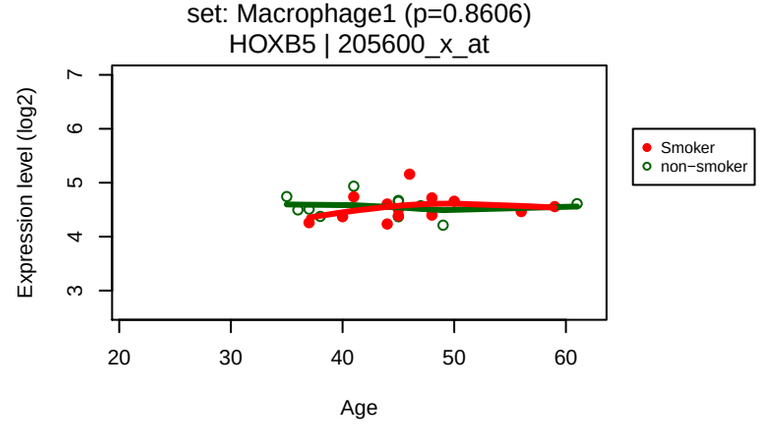

set: SmallAirway1 (p=0.6293)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker

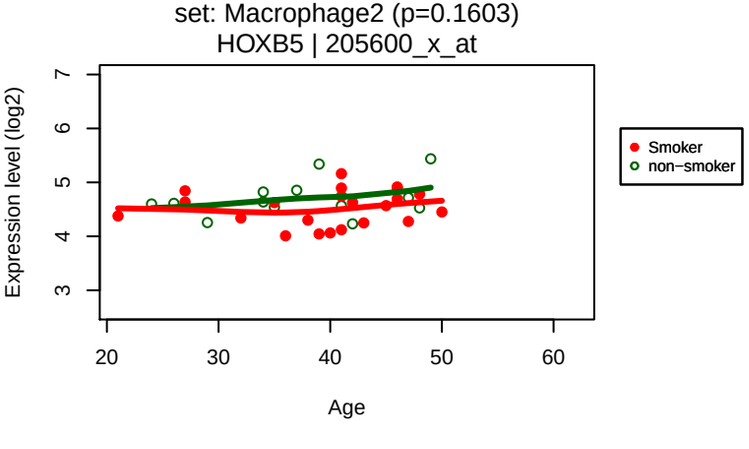

set: SmallAirway2 (p=0.7208)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker

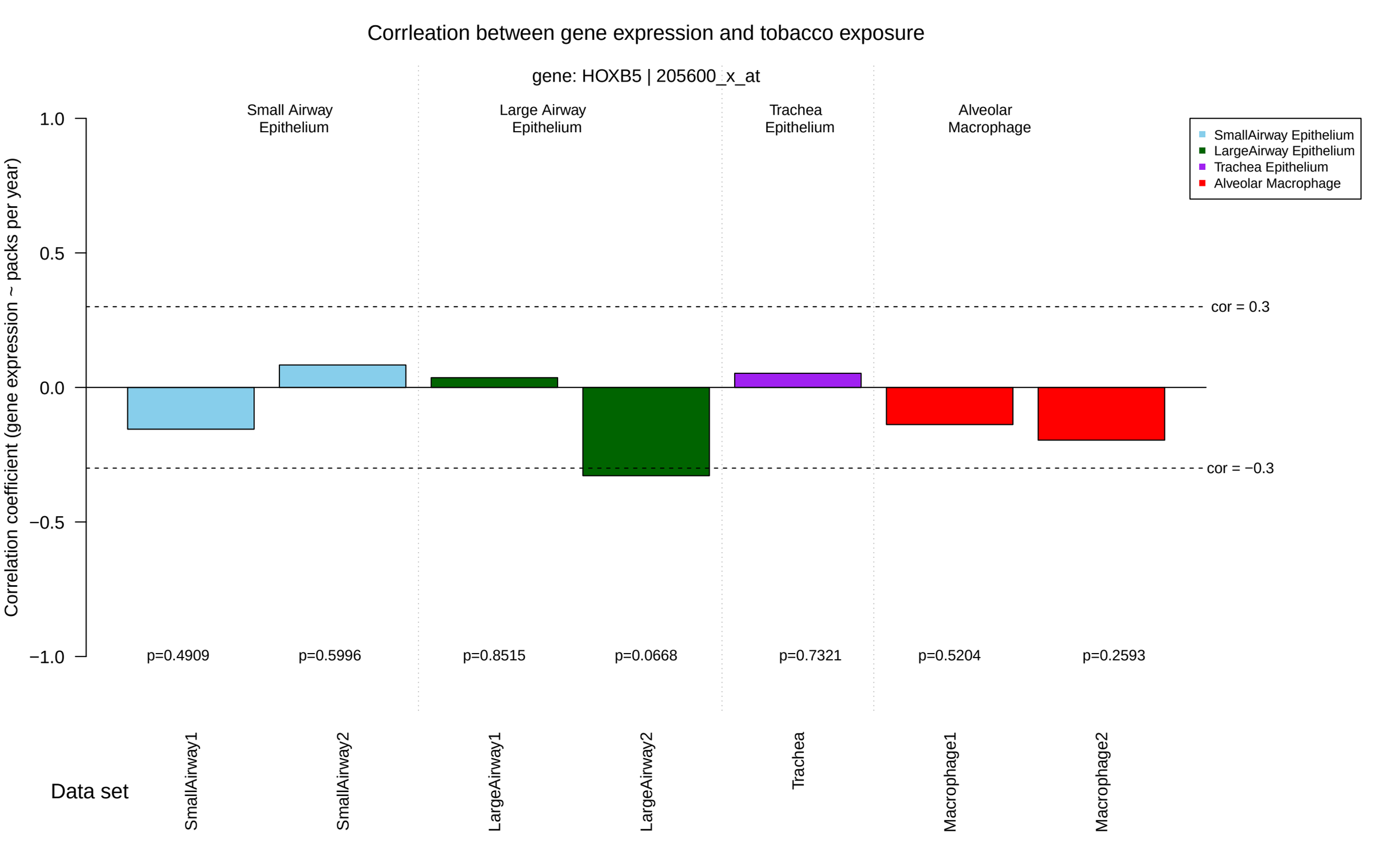

set: LargeAirway1 (p=0.2386)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1673)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2989)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8606)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1603)
HOXB5 | 205600_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB5 | 205600_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4909
p=0.5996
p=0.8515
p=0.0668
p=0.7321
p=0.5204
p=0.2593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2